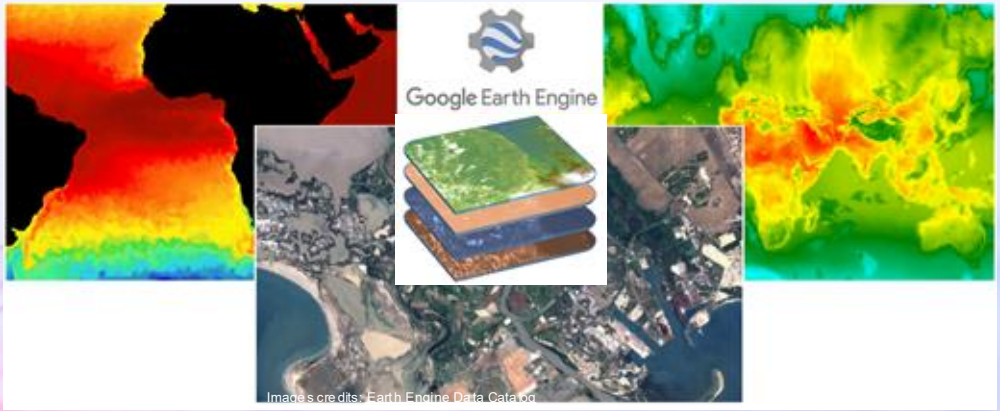
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GEOAI FUSION 2026

[Geospatial Analysis in Google Earth Engine (GEE) and Machine Learning Concepts in GEE]

[Assoc. Prof. Ts. Gs. Dr Wan Shafrina Wan Mohd Jaafar] | [National University of Malaysia (UKM)]

[21 July 2026] • [Bootcamp 2 / Module 5] • ASEAN GeoAI Fusion 2026

[Session 02] • Opening Lecture • 9:05 – 9:35 am



Session Agenda

SECTION 01

Why Google Earth Engine? and what is it?

SECTION 02

GEE in Research

SECTION 03

Communicating Results with GEE Apps

SECTION 04

2026 Geospatial Landscape

SECTION 05

What is covered in this module

Why Google Earth Engine? and what is it?

Cloud-based planetary-scale geospatial analysis — without downloading a single file


OSTM/Jason 2
(NOAA)
Aquarius
QuikSCAT
TRMM
Terra
ISS-RapidScat,
CATS
EO-1
Landsat 7
(USGS)
SMAP
Aqua
Suomi NPP
(NOAA)
Landsat 8
(USGS)
SORCE,
TCTE (NOAA)
GPM
Aura
GRACE (2)
CALIPSO
CloudSat
OCO-2

Source: NASA

> 50 Petabytes of Earth observation data (imagery, weather, etc.)



Source: Earth Engine Team
https://goo.gl/n5Gh5Q

Massive archive of Public Earth observation data

SATELLITE IMAGERY

90+

petabytes of data, ready to query

1,000+

datasets across satellites, climate & vectors

50+ years

Landsat archive begins — 50+ years of history

daily update

new imagery added continuously

- Landsat 4–9 · 30m · 1982–present
- Sentinel-1 & 2 · 10m · 2014–present
- MODIS · 250m–1km · 2000–present
- SRTM / NASADEM elevation

- ERA5 — hourly global climate reanalysis
- CHIRPS — daily rainfall · 1981–present
- NLCD, MODIS land cover — global LULC
- TIGER — US census boundaries & roads

Massive archive of Community Data

Awesome GEE Community Catalog

Data Themes

- Population & Socioeconomic
- Geophysical, Biological & Biogeochemical
- Elevation and Bathymetry
- Soil Properties
- Global Land Use and Land Cover
- Regional Land Use and Land Cover
- Hydrology
- Oceans and Shorelines
- Agriculture, Vegetation and Forestry
- Analysis Ready Data
- Global Utilities, Assets and Amenities Layers
- Biodiversity, Ecosystems & Habitat Layers
- Weather and Climate Layers
- Global Events Layers
- Fire Monitoring and Analysis
- Insiders only datasets

Google Earth Engine: geospatial analysis at planetary scale without downloading a single file

Traditional workflow

- Download satellite scenes — GBs per image
- Preprocess locally: cloud mask, reproject, mosaic
- Run analysis on your laptop — one region at a time
- Share results as large raster files
- Colleagues need your code, environment, and data

GEE workflow

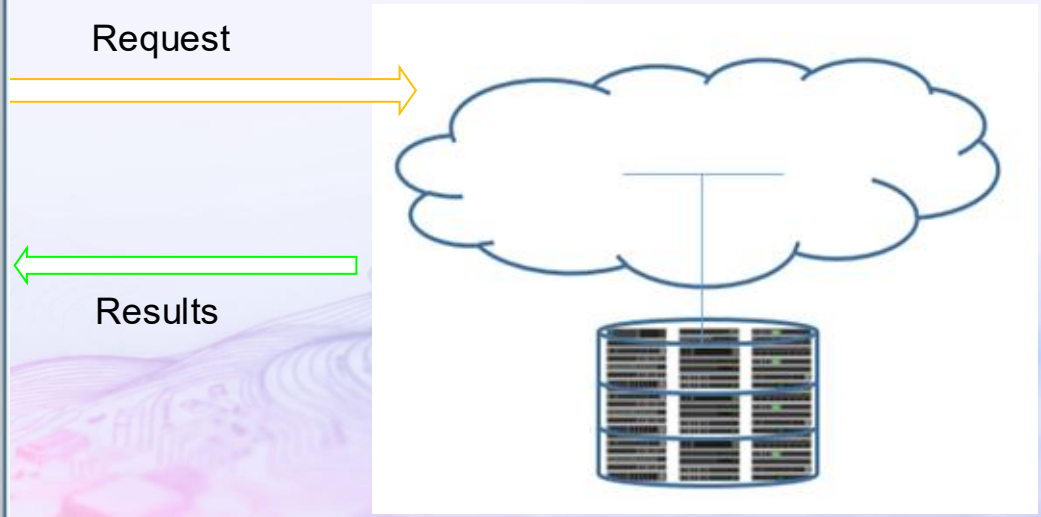
- Data lives on Google's servers — already there
- Write code in a browser — runs in the cloud
- Scale from one county to the planet — same code
- Share a link — anyone runs it, no install needed
- Free for research and education

What used to take a computing cluster and weeks of preprocessing can run in minutes in a browser.

Client writes. Server computes. You see results.

CLIENT-SIDE (YOUR BROWSER)	SERVER-SIDE (GOOGLE'S MACHINES)	CLIENT-SIDE (YOUR BROWSER)	
Your browser	**Google servers**	**Map + Console**	**Export / Share**
You write JS code in the Code Editor	ee. objects execute here, on petabytes	Results render as tiles in your browser	Files to Drive, Cloud, or GEE Assets



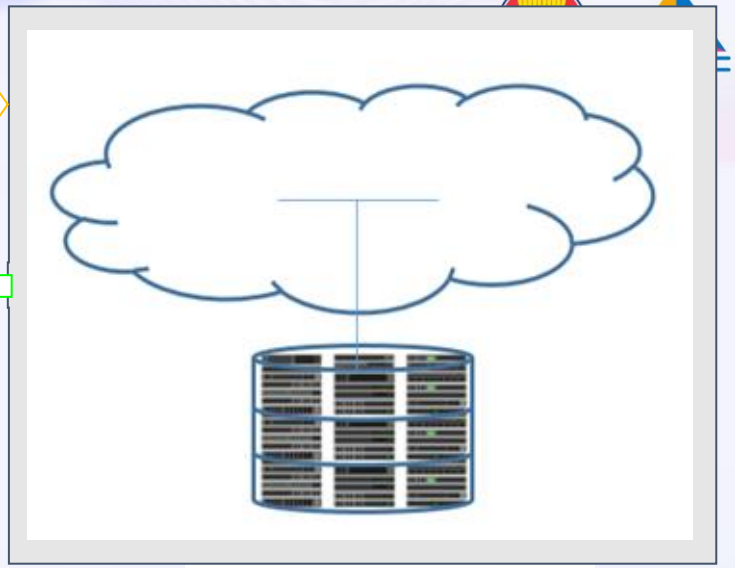




Original Image's credits: Pixel.com

Request

Results



- The JavaScript API using the code editor is widely used to interact with GEE

JavaScript

- Another option to access is using Google colab for Python API

Python

From a single script — analyses that used to take months

IMAGE ANALYSIS

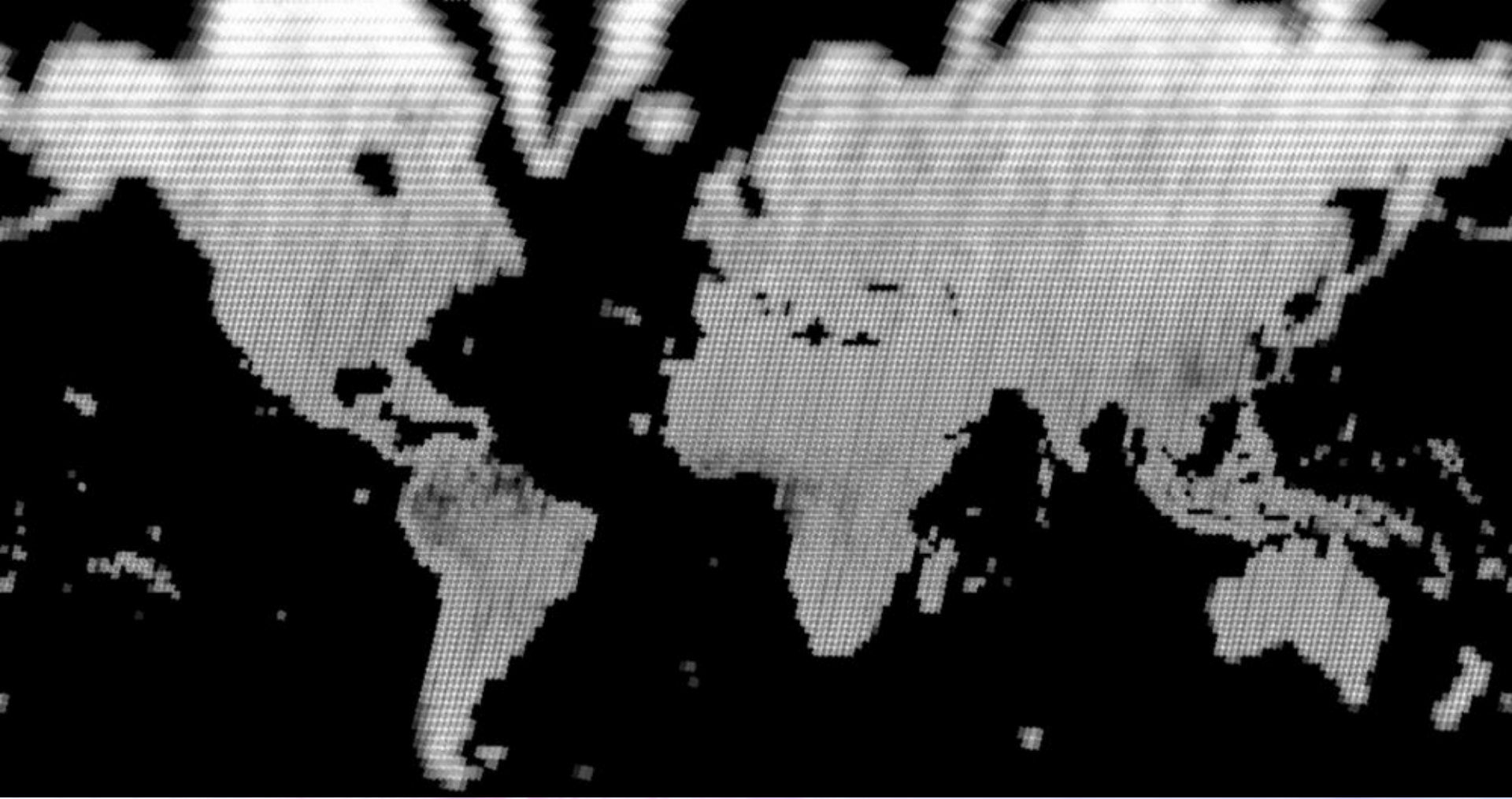
- Spectral indices — NDVI, LST, NDWI, NBR
- Cloud-free composites over any region
- Land cover classification
- Change detection between any two dates
- Export Images

TIME SERIES

- 50-year vegetation or temperature trends
- Annual, seasonal, or monthly aggregation
- Interactive charts in the browser
- Export time series tables as CSV

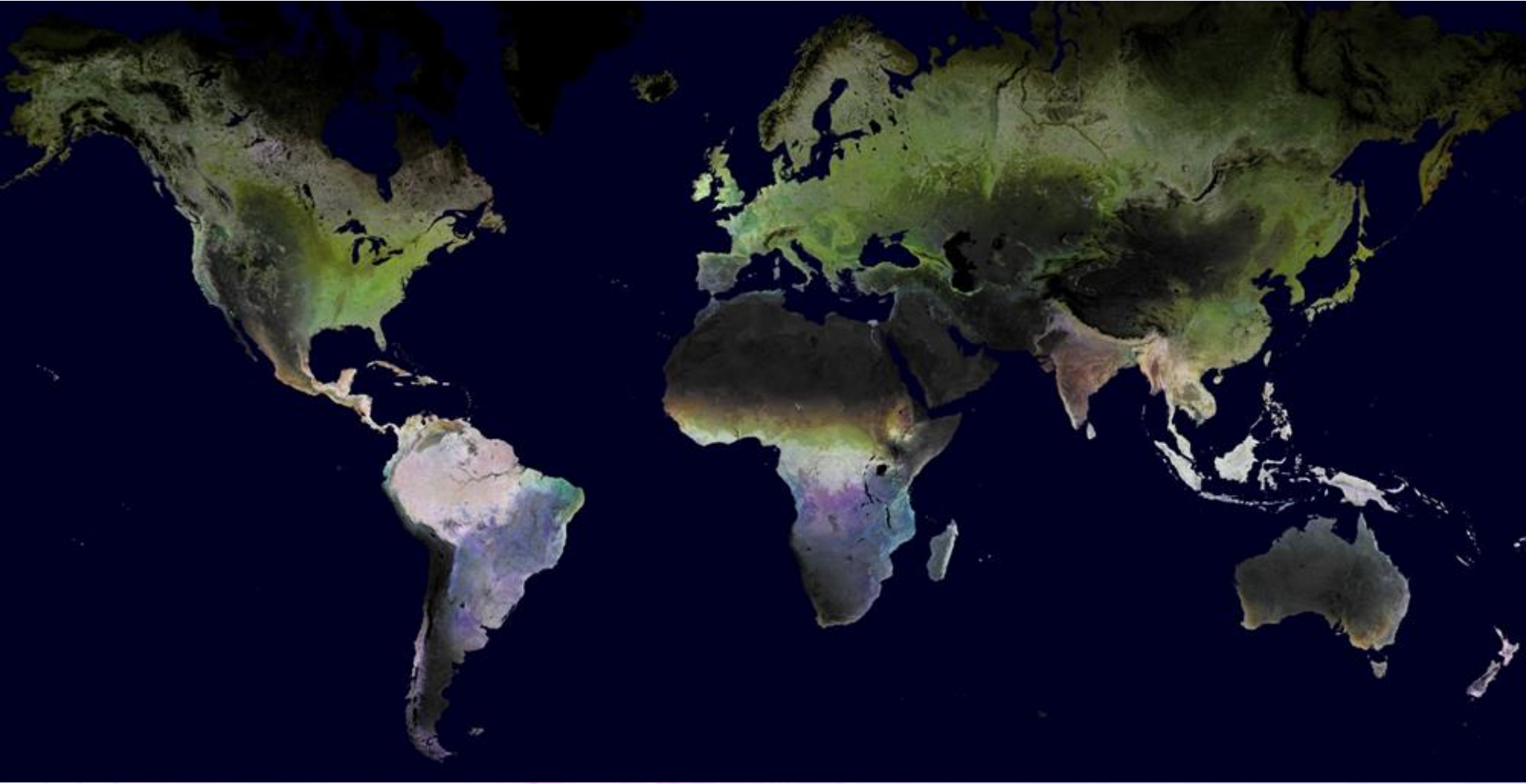
VECTORS & TABLES

- Zonal statistics by county or watershed
- Area calculations at any scale
- Join raster values to survey points
- Export GeoJSON or CSV for R/Python



BIG data: 40+ years of Landsat

Source: Earth Engine Team
https://goo.gl/n5Gh5Q



Daily MODIS composites

Source: Earth Engine Team
https://goo.gl/n5Gh5Q


1984
Source: Earth Engine Team
https://goo.gl/n5Gh5Q


1984
https://earthengine.google.com/timelapse/

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MODIS NDVI Times Series Animation

Link



Source: GEE Community Tutorials Link

GEE in Research

Landmark studies · application domains · real-world operations

ELSEVIER

Remote Sensing of Environment

Volume 202, 1 December 2017, Pages 18-27



Google Earth Engine: Planetary-scale geospatial analysis for everyone

Noel Gorelick [a], Matt Hancher [b], Mike Dixon [b], Simon Ilyushchenko [b], David Thau [b], Rebecca Moore [b]

GEE has reshaped how environmental science is done

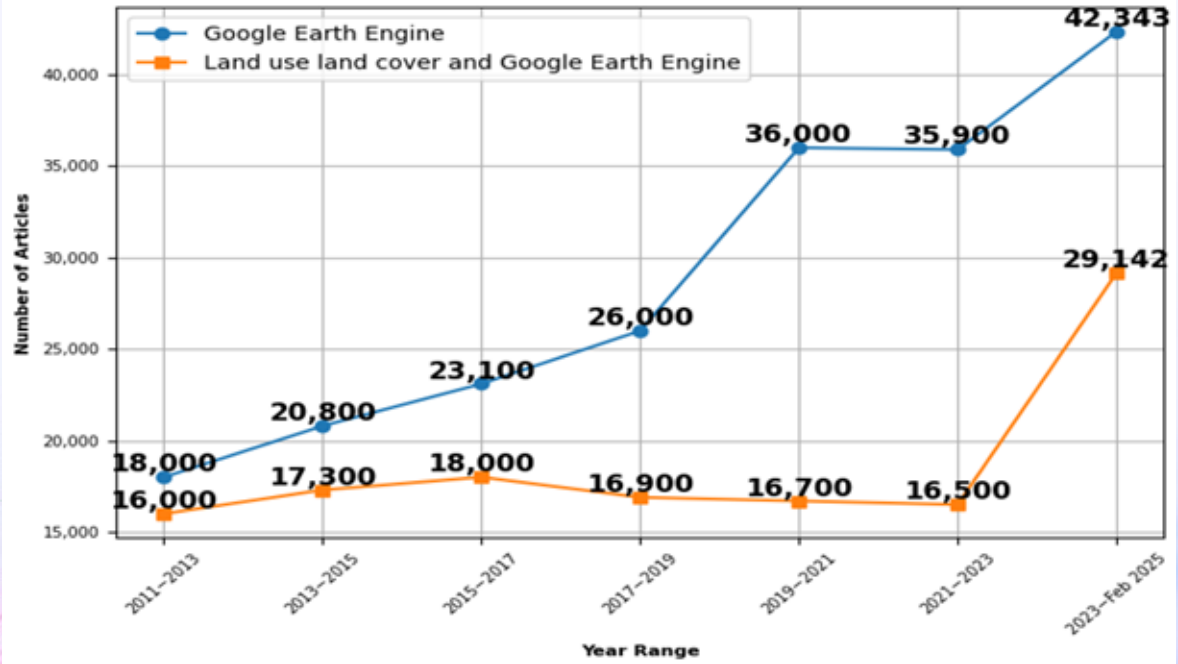
Gorelick et al. 2017, Remote Sensing of Environment — the most cited RS platform paper ever

16,000+

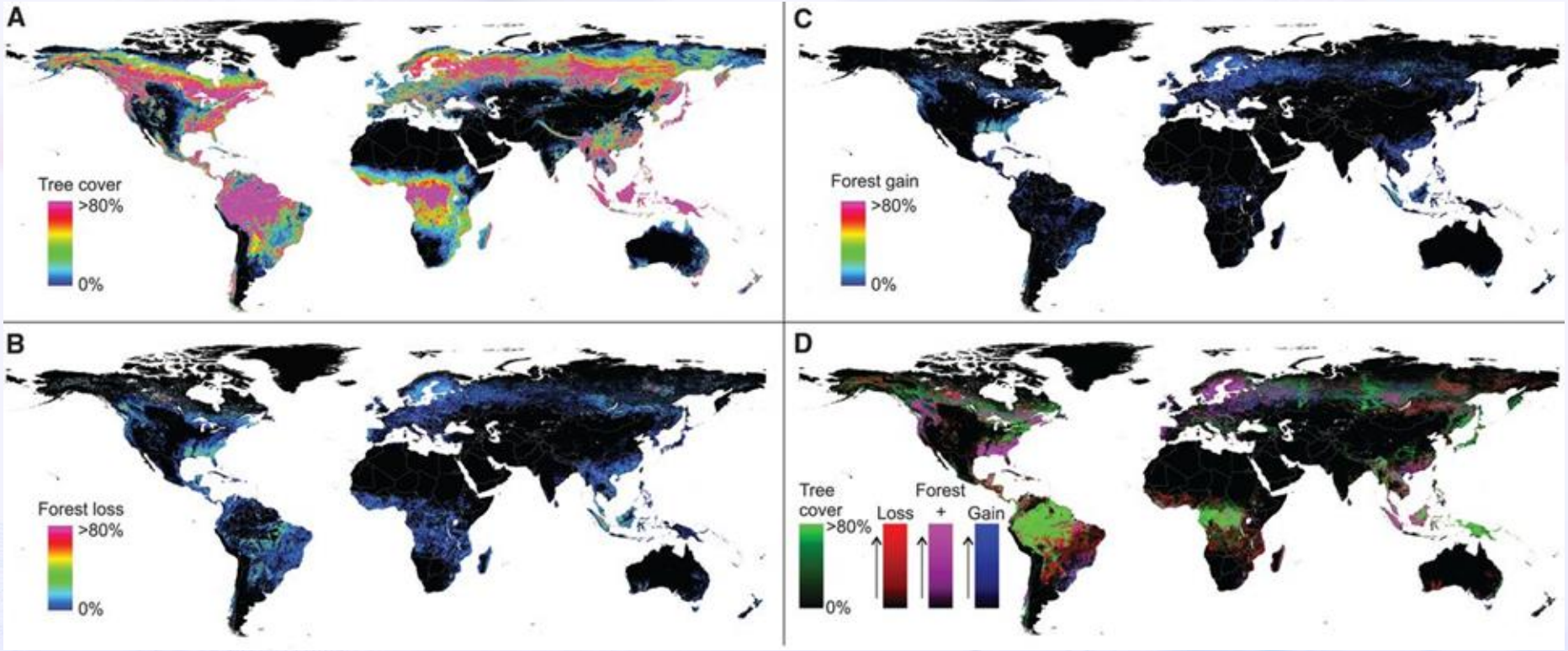
citations of Gorelick et al. 2017

28,000+

studies from 125 countries



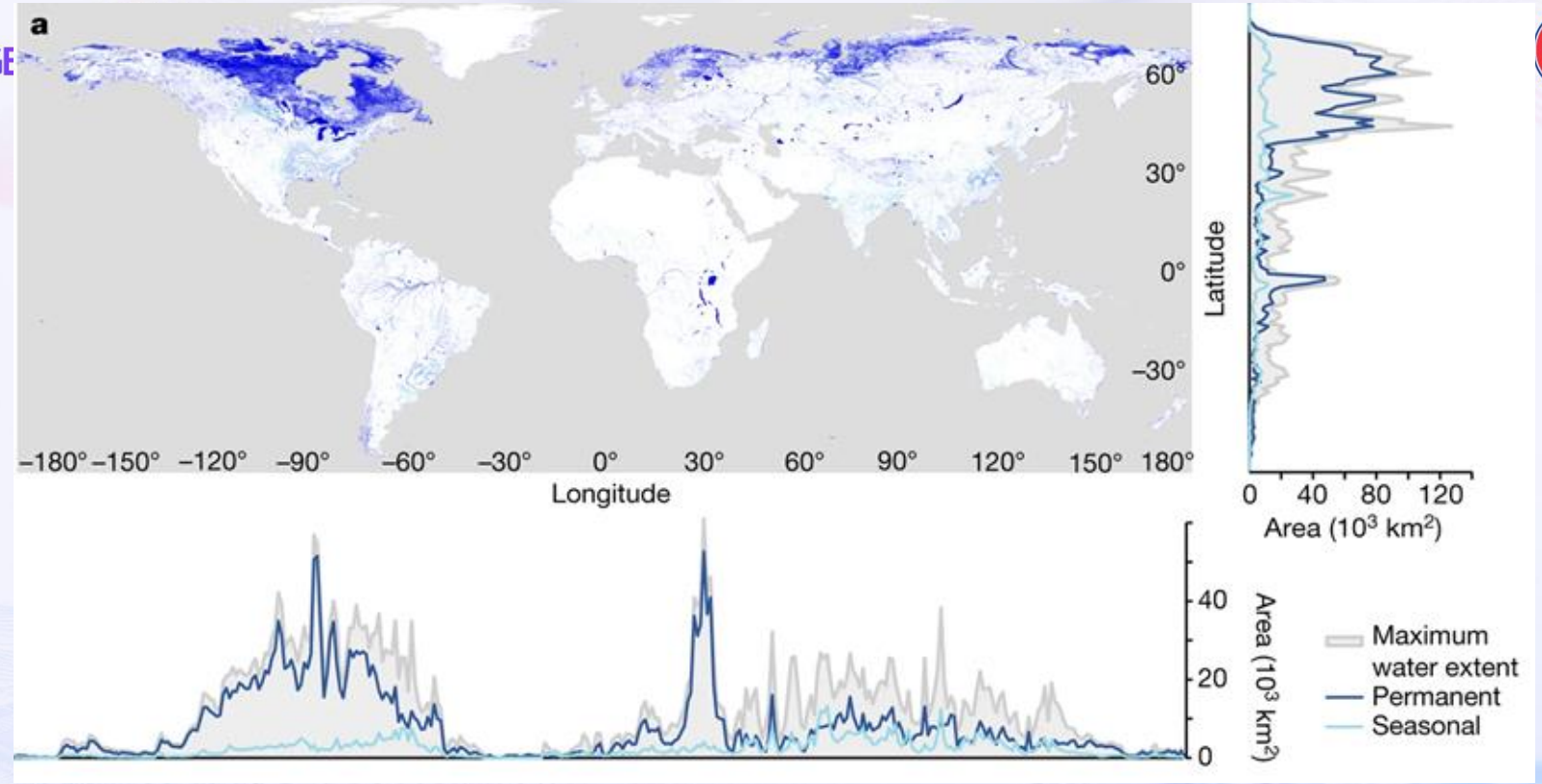
Source: Alshehri et al 2025



High-Resolution Global Maps of 21st-Century Forest Cover Change

Hansen et al. (2013) · Science

Processed 3+ million Landsat images to produce the first 30m global forest change map. Revealed that Indonesia surpassed Brazil in rainforest loss in 2012. Powers Global Forest Watch — used by governments and NGOs worldwide.

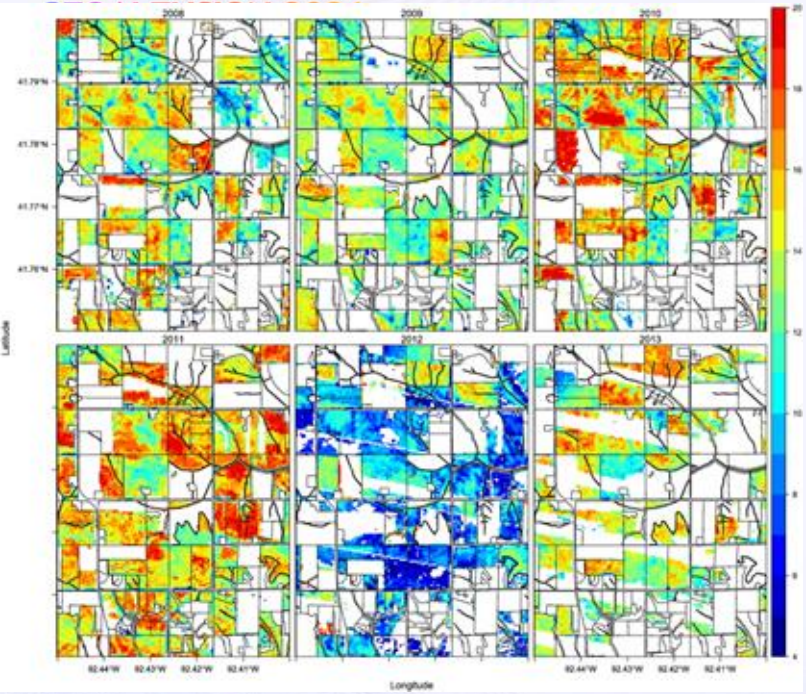


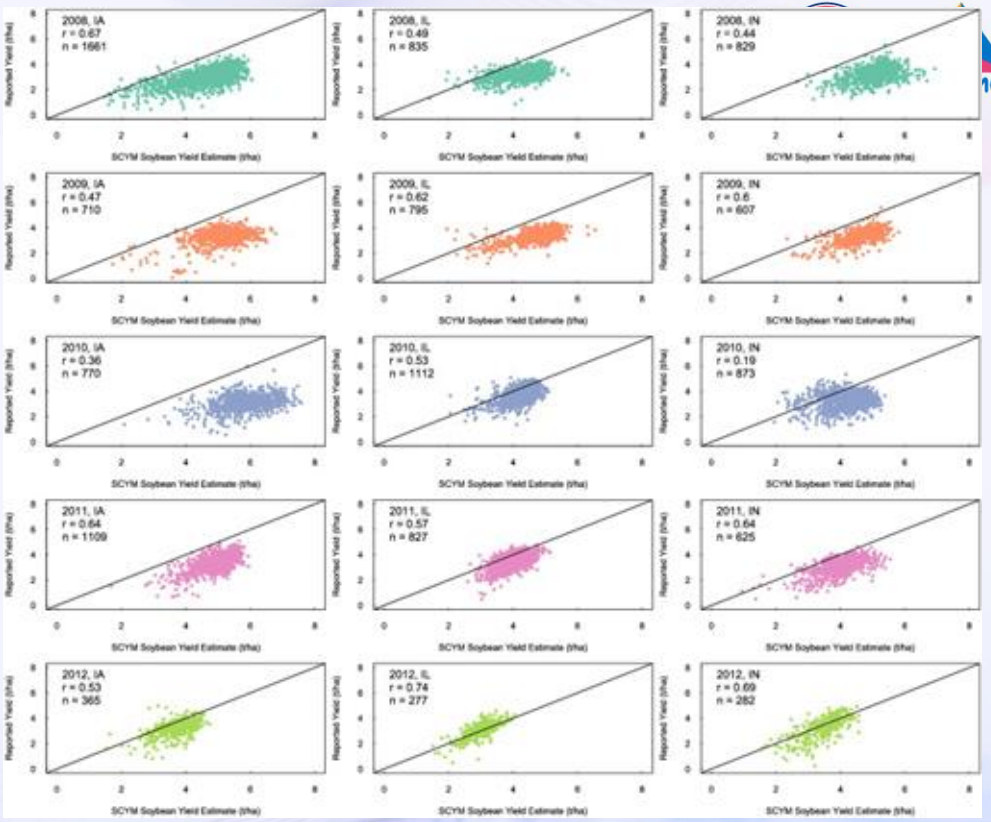
High-resolution mapping of global surface water and its long-term changes

Pekel, Cottam, Gorelick & Belward (2016) · Nature

Analysed 3 million+ Landsat images — surface water change at 30m over 32 years. Found surface waters declined by ~90,000 km², largely in the Middle East and Central Asia. The JRC dataset is now one of the most-used in GEE.

Asian Countries Earth Analytics





A scalable satellite-based crop yield mapper

Lobell et al. (2015) · Remote Sensing of Environment

Used GEE to map crop yields at field scale across the US Corn Belt — previously impossible at this resolution. Demonstrated satellite data + cloud computing could replace expensive ground surveys for food security monitoring.



Source: Tamiminia et al 2020

GEE is used across nearly every field that touches land, water, or atmosphere

Bibliometric analysis of 2,800+ studies · Velastegui-Montoya et al. 2023; Tamiminia et al. 2020

FORESTS & LAND COVER

- Global deforestation monitoring
- LULC change detection
- Mangrove and wetland mapping
- Burn severity assessment

AGRICULTURE & FOOD

- Crop type classification
- Yield estimation at field scale
- Paddy rice mapping
- Smallholder farm detection

CLIMATE & ATMOSPHERE

- LST trend analysis
- Evapotranspiration estimation
- Snow cover and glacial retreat
- CO_2/CH_4 emission hotspots

WATER & HYDROLOGY

- Surface water occurrence (JRC)
- Wetland inundation
- Flood extent mapping
- Drought indices (NDWI, SPI)

URBAN & INFRASTRUCTURE

- Impervious surface expansion
- Urban heat island mapping
- Building footprint extraction
- Transportation network analysis

HEALTH & DISASTER

- Malaria risk mapping (Sturrock et al. 2014)
- Disaster damage assessment
- Air quality proxies
- Flood displacement estimation

Communicating Results with GEE Apps

From analysis script to shareable web app — one button

Your analysis is done. Now how do you share it?

TRADITIONAL OPTIONS

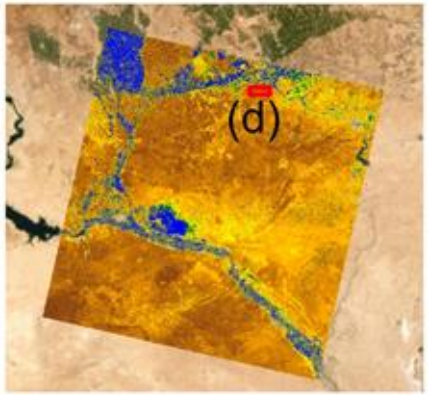
- Export a static map image
- Share a GeoTIFF — needs GIS software to open
- Send a figure in a PDF — no zoom, no query
- ArcGIS Online / QGIS Cloud — separate platform, fees
- Colleagues need your code, environment, and data

GEE APPS

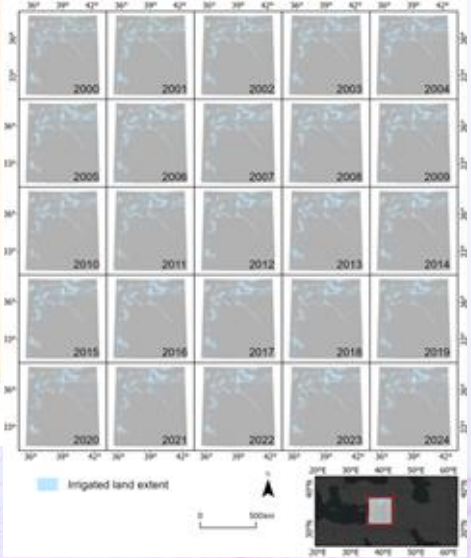
- Publish your script as a web app
- Anyone with the link can explore it — no GEE account
- Interactive — zoom, click, change parameters
- Always live — updates when you update the script
- Shareable to public URL or a Google Group

Your analysis is done. Now how do you share it?

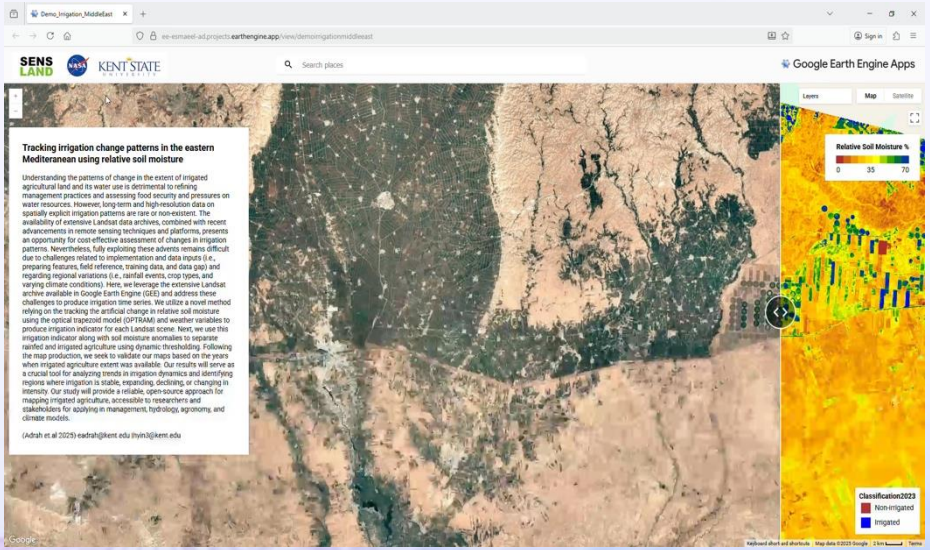
TRADITIONAL OPTIONS

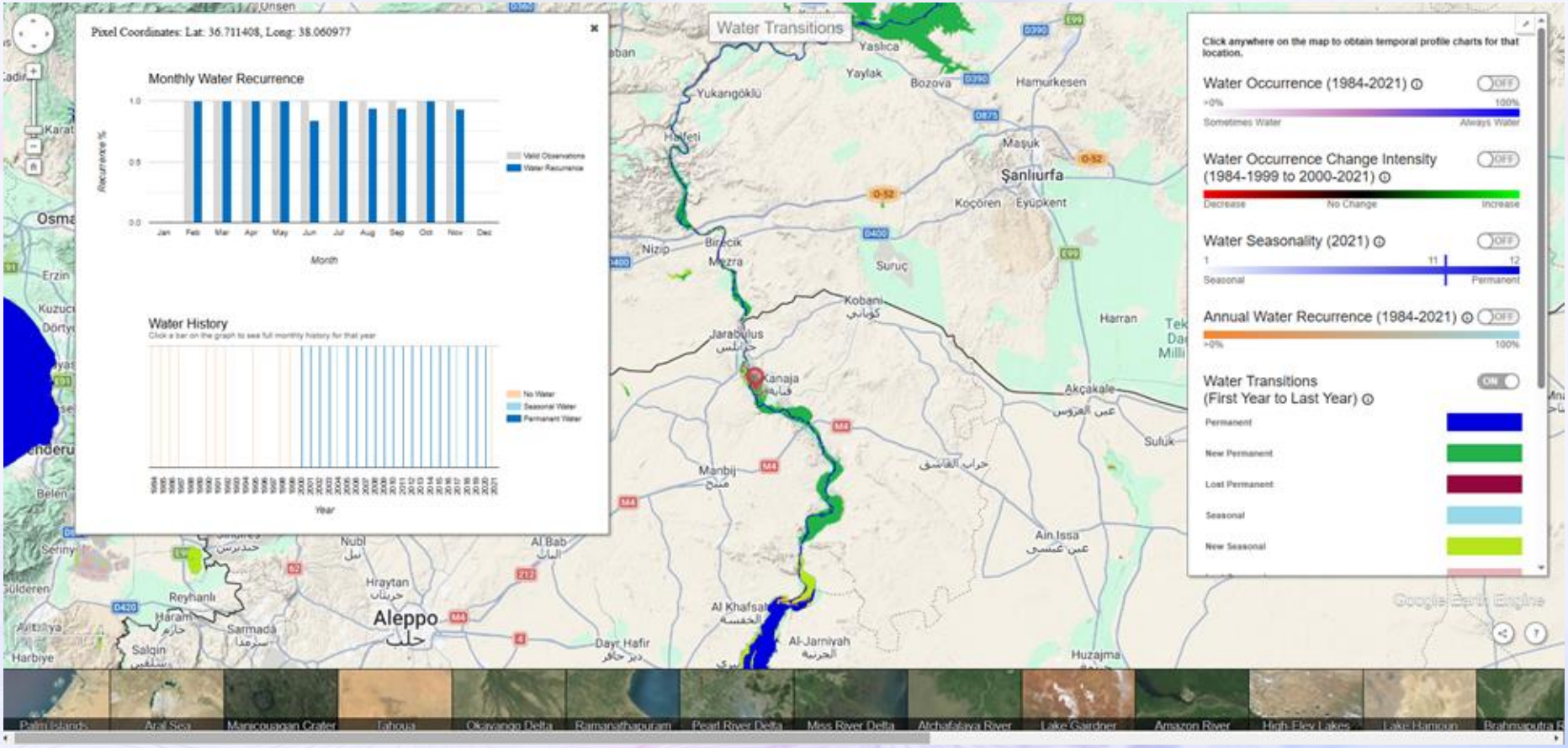


Landsat image May 11th 2016 - Footprint (WRS path 172, WRS row 35)

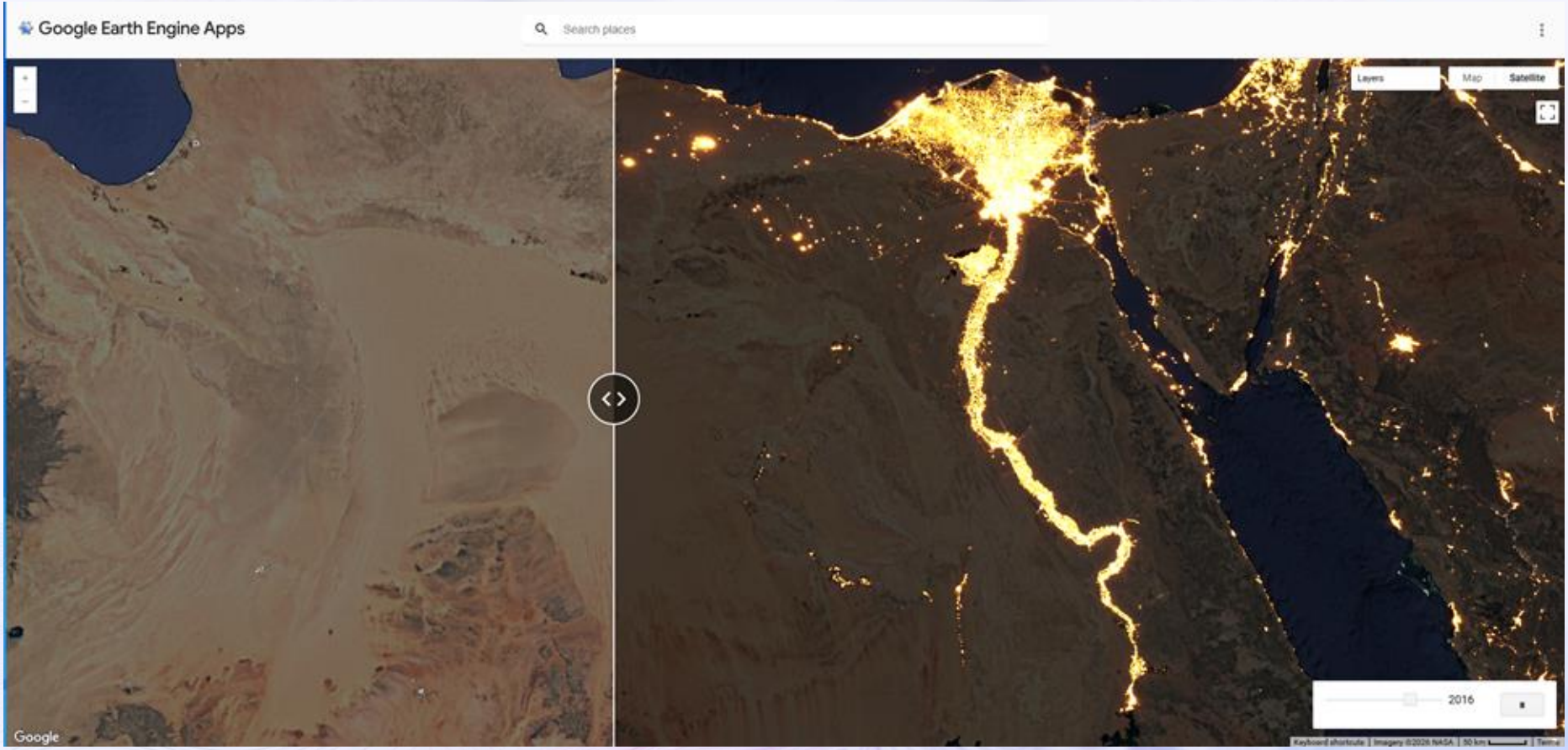


GEE APPS





https://global-surface-water.appspot.com/map



https://gena.users.earthengine.app/view/urban-lights

Climate Engine Beta

Guided Tours | Video Tutorials | Examples | Manual | Contact

ClimateEngine | Data | Metrics | References | Sponsors

Map Layer Options | Time Series Options

GET MAP LAYER

Product

Type: Climate

Dataset: UI METDATA/gridMET

Variable: ETo (ASCE Grass Reference Evapotranspiratic

Processing

Calculation: Percent Difference From Average Conditions

Statistic: Mean

Time Period

(Data:1979-01-01 to 2016-09-25)

Last Year

Start Date: 2015-09-26

End Date: 2016-09-25

Year Range for Historical Avg/Distribution: 1981 - 2010

TOGGLE MENU | Map | Animations

Colormap Options | Apply Mask | Download | Get Link | Reset Forms

Mean Reference Evapotranspiration Percent Difference from Average

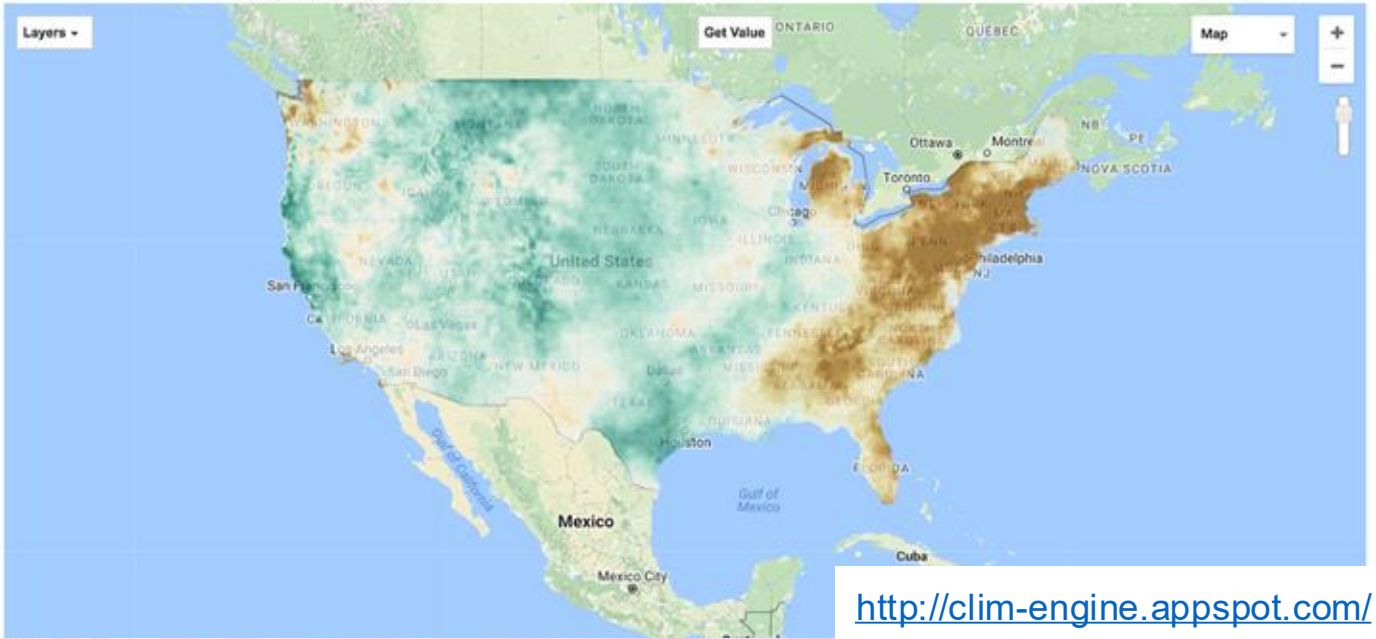
Data Source: METDATA/gridMET 4-km dataset (University of Idaho)

Target Period: 2015-09-26 to 2016-09-25

Historical Period: 1981 - 2010

-20 -15 -10 -5 0 5 10 15 20

Reference Evapotranspiration Percent Difference from Average (%)



http://clim-engine.appspot.com/

GEE Workshop in 2026 Geospatial Landscape!

Is coding still matters in 2026, since LLMs can do it.
What about Geospatial Foundation Models?

Can we go through a typical “hello world” tutorial and ignore the elephant in the room

Large language Models

Geospatial Foundation Models

LLMs can already write GEE code…

WHAT LLMS CAN DO TODAY

- First draft of GEE scripts from your description
- Turning analysis code into interactive apps
- Explaining an unfamiliar error message
- Adapting an example

WHAT THEY STILL STRUGGLE WITH

- Know if the output is geographically correct
- Understand your research question
- Catch a wrong dataset failing silently
- Decide what analysis is worth doing
- Validate results

The question is not should I use LLMs? — it's how do I use them without losing the ability to judge their output?

LLM and The skill ladder problem

Experts became experts by struggling through fundamentals. What happens when that struggle is skipped?

The risk is not that LLMs replace experts — it's that fewer people develop into experts in the first place.

Struggle in a new way, don’t skip the fundamentals

USES LLM INTENTIONALLY

- First draft of GEE scripts from your description
- Turning analysis code into interactive apps
- Explaining an unfamiliar error message
- Adapting an example to your study area
- Writing UI or specific functions

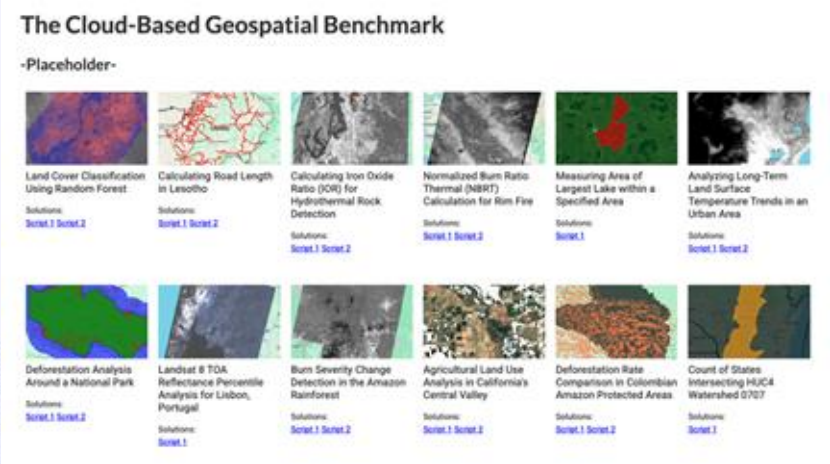
Harness your domain expert

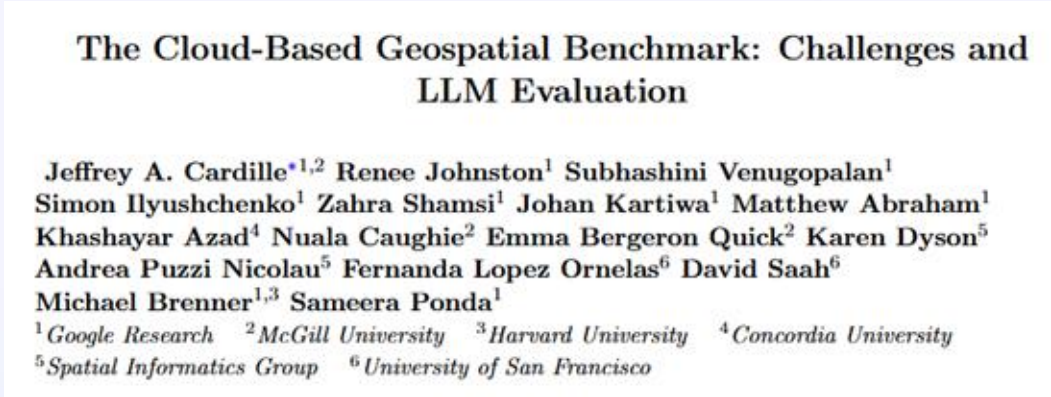
- Defining the right questions to ask LLM
- Interpreting what the map actually shows
- Deciding which dataset is appropriate
- Critically scrutinize outputs against known values and your domain knowledge **(Designing TEST and Connecting results to real world)**

How can you trust the results?

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Evaluating LLM generated code in Earth Engine



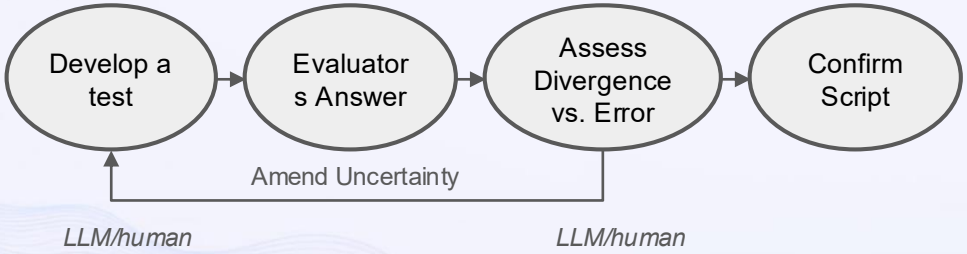


The Cloud-Based Geospatial Benchmark: Challenges and LLM Evaluation

Jeffrey A. Cardille[*1,2] Renee Johnston[1] Subhashini Venugopalan[1]
Simon Ilyushchenko[1] Zahra Shamsi[1] Johan Kartiwa[1] Matthew Abraham[1]
Khashayar Azad[4] Nuala Caughie[2] Emma Bergeron Quick[2] Karen Dyson[5]
Andrea Puzzi Nicolau[5] Fernanda Lopez Ornelas[6] David Saah[6]
Michael Brenner[1,3] Sameera Ponda[1]
[1] *Google Research* [2] *McGill University* [3] *Harvard University* [4] *Concordia University*
[5] *Spatial Informatics Group* [6] *University of San Francisco*

Google Earth Engine EE open source textbook (eefabook.org).

Evaluating LLM generated code in Earth Engine



What must be clear in a useful question?

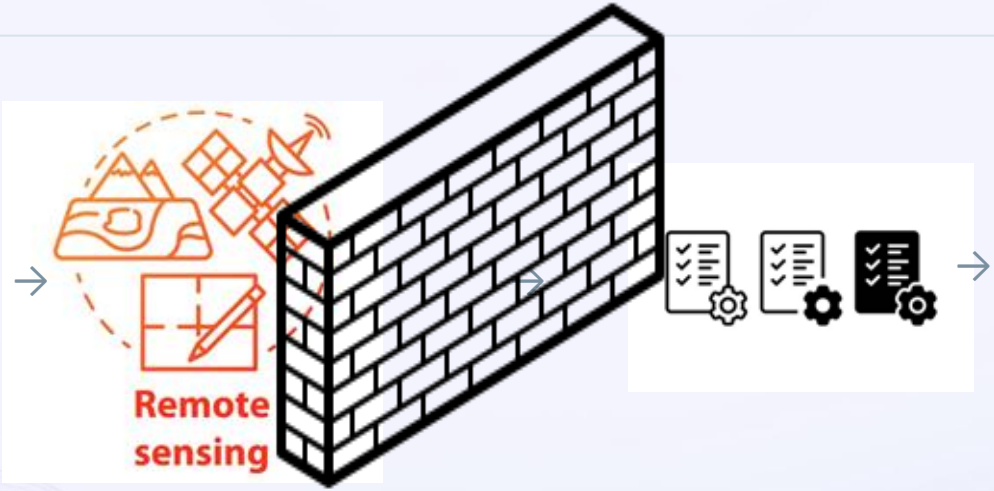
1	Data specified
2	Data exists
3	Dates good
4	Data needed
5	Latest versions
6	Nothing deprecated
7	Invented properly
8	Labeled if invented
9	Lon/Lat
10	Points well placed
11	Co-located
12	Timing precise
13	Boundaries clear
14	Analysis resolution

15	Seed Provided
16	Cloud Treatment
17	Image threshold
18	Pixel masking
19	Reduction explicit
20	Statistics clear
21	No hints
22	No pitfalls
23	No slang
24	Wording clear
25	Not wordy
26	Clear answer
27	Single answer
28	Units clear

Source: Cardil et al 2025

Geospatial Foundation Models

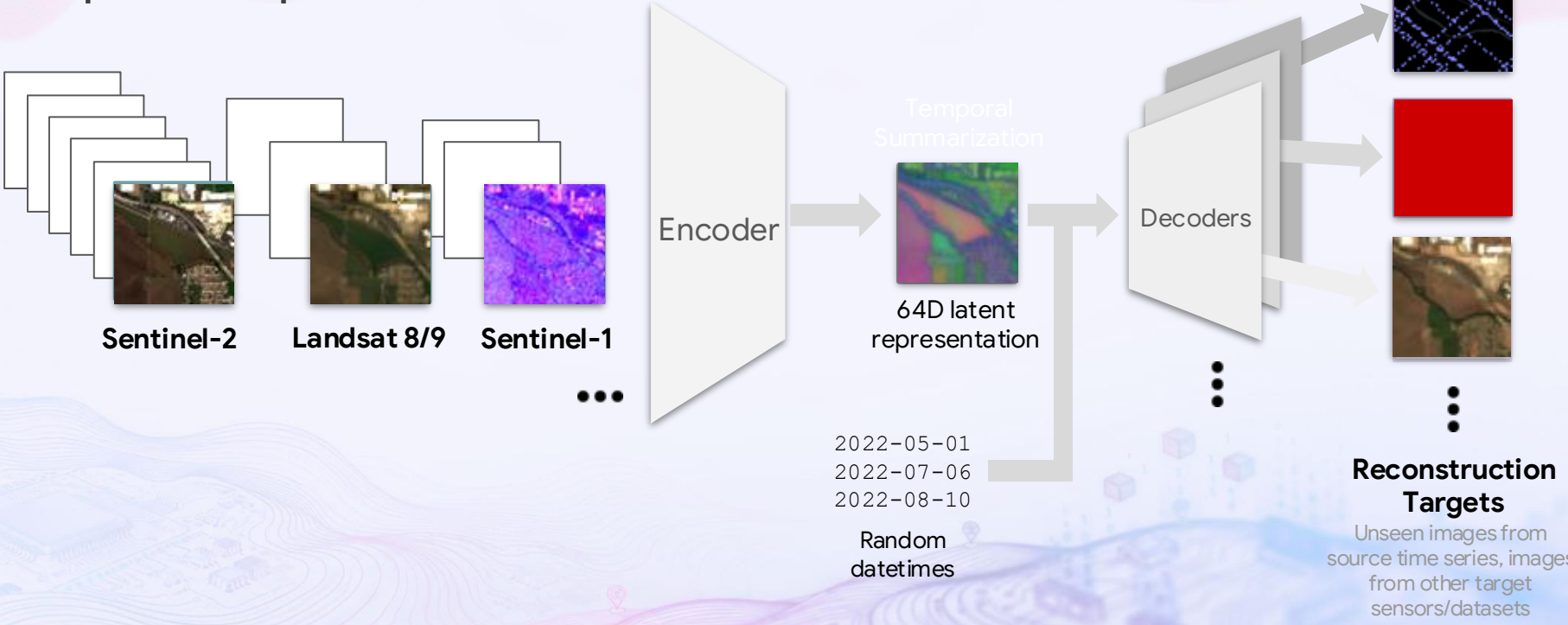




Raw Data → Expert Knowledge (Remote sensing & processing) → Extracted Features (Analysis ready data) → Domain Expert Analysis → Answers

Foundation Models → Embeddings → Domain Expert Analysis → Answers

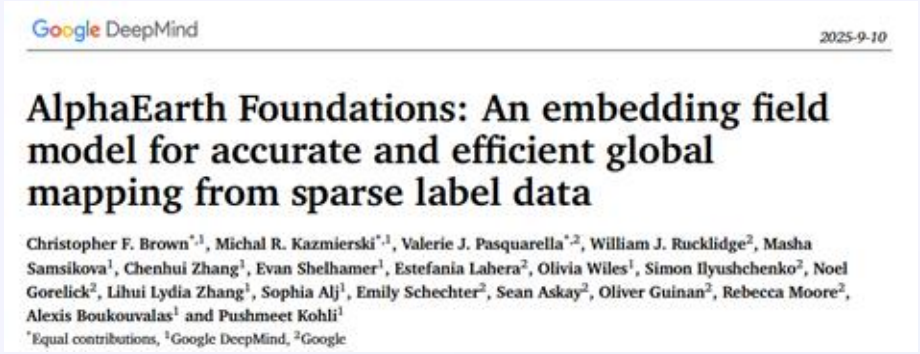
Super-simplified overview



Source: Pasquarella et al & Google Team 2026
https://goo.gle/geoai-in-ee-itu-2026

Alpha Earth Foundations & Satellite Embeddings

- AlphaEarth Foundations was designed to function as a **virtual satellite** whose images summarize Earth's surface dynamics over space and time
- Learns **meaningful patterns** across data streams and generate spatially and temporally precise embedding vectors for any place, any time
- Outputs were designed for **practitioners** – like an AI-powered image composite
- **Embedding images** are a foundation on which many different EO applications can be built



Google DeepMind 2025-9-10

AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data

Christopher F. Brown*,1, Michal R. Kazmierski*,1, Valerie J. Pasquarella*,2, William J. Rucklidge2, Masha Samsikova1, Chenhui Zhang1, Evan Shelhamer1, Estefania Lahera2, Olivia Wiles1, Simon Ilyushchenko2, Noel Gorelick2, Lihui Lydia Zhang1, Sophia Alj1, Emily Schechter2, Sean Askay2, Oliver Guinan2, Rebecca Moore2, Alexis Boukouvalas1 and Pushmeet Kohli1
*Equal contributions, 1Google DeepMind, 2Google



Satellite Embedding V1

Dataset Availability
2017-01-01T00:00:00Z - 2024-01-01T00:00:00Z

Dataset Provider
Google Earth Engine Google DeepMind

Earth Engine Snippet

Description Bands Image Properties Terms of Use

The Google Satellite Embedding dataset is a global, analysis-ready collection of learned geospatial embeddings. Each 10-meter pixel in this dataset is a 64-dimensional representation, or "embedding vector," that encodes temporal trajectories of surface conditions at and around that pixel as measured by various Earth observation instruments and datasets, over a single calendar year. Unlike conventional spectral inputs and indices, where bands reflect physical measurements, embeddings are feature vectors that summarize relationships across multi-source, multi-modal observations in a less directly interpretable, but more powerful way.

Source: Pasquarella et al & Google Team 2026
https://goo.gle/geoai-in-ee-itu-2026



Source: https://earthengine-ai.projects.earthengine.app/view/satellite-embedding-rgb#red=A26;green=A04;blue=A22

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Key Resources

Data Catalog & Community Catalog	GEE data catalog: https://developers.google.com/earth-engine/datasets GEE data community catalog: https://gee-community-catalog.org/
GEE Official Documentation & Community Tutorials	GEE documentation: https://developers.google.com/earth-engine/guides/getstarted Community Tutorials: https://developers.google.com/earth-engine/tutorials
GEE Open source textbook	The book: https://www.eefabook.org/; The repository: https://goo.gle/EEFA_repo
Supplementary	GEE Chrome Extension; LLM assisted coding; Open Earth Engine Library;
Integration Python API - geemap	Python Installation page: https://developers.google.com/earth-engine/guides/python_install geemap library: https://geemap.org/ QGIS Plug in: Plugin repository: qgis.gishub.org GitHub:https://github.com/opengeos/qgis-timelapse-plugin
This workshop Repository	This workshop repository: https://code.earthengine.google.com/?accept_repo=users/shafrina__geoai/bootcamp2

Thank You

Questions & Discussion

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 [LinkedIn/wan-shafrina-wan-mohd-jaafar]

 [UKMProfile/Wan_Shafrina